

MATH 521

Real Analysis

Final Practice Book

Write-On Workbook

University of Wisconsin–Madison

Spring 2026

Step 1 build: Homework 9

Name: _____

Date: _____

Contents

How to Use This Workbook	ii
1 Homework 9	1
1.1 Problem 1	2
1.2 Problem 2	12
1.3 Problem 3	22
1.4 Problem 4	32
1.5 Problem 5	41
1.6 Problem 6	51
2 Homework 10	60
2.1 Problem 1	61
2.2 Problem 2	70
2.3 Problem 3	79
2.4 Problem 4	89
3 Homework 11	98
3.1 Problem 1	99
3.2 Problem 2	109
3.3 Problem 3	119
3.4 Problem 4	127
3.5 Problem 5	135
3.6 Problem 6	143
4 Homework 12	153
4.1 Problem 1	154
4.2 Problem 2	164
4.3 Problem 3	174
4.4 Problem 4	184
4.5 Problem 5	194
4.6 Problem 6	204

How to Use This Workbook

This workbook is for active proof practice. For each problem, work through the diagnosis, recipe card, self-explanation, skeleton, and full proof attempt before reading the reference solution.

- Pass 1** Diagnose the topic, task, relevant definitions, and likely proof method.
- Pass 2** Convert the problem into hypotheses, conclusion, tools, and a proof plan.
- Pass 3** Answer targeted questions that expose the proof's logic.
- Pass 4** Reconstruct the proof from a skeleton without reading the full solution.
- Pass 5** Write a complete proof from scratch in the blank writing space.

Only after Pass 5 should you compare your work to the official or reference solution.

Homework 9

Problem 1

Problem Statement

Homework 9, Problem 1

Consider the metric space $(\mathbb{R}, d)$ with the usual metric $d(x, y) = |x - y|$. For each of the following sets $A \subseteq \mathbb{R}$, find its closure, isolated points, and limit points. Prove your answer.

- (a) $A = [0, 1)$
- (b) $A = \mathbb{Z}$
- (c) $A = \{x \in \mathbb{Q} : 0 < x < 1\}$

Topic: _____

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

1. For each set, which points are obvious members of the closure, and which boundary points require a ball argument?

2. How do you prove that a point is not in $\overline{\mathbb{Z}}$? What radius separates it from all integers?

3. What is the difference between proving $a \in A$ is isolated and proving $a \in A$ is a limit point?

4. Where is density of $\mathbb{Q}$ used in part (c), and why is it not enough just to say “there are rationals nearby”?

5. How does the identity $\overline{A} = (\text{isolated points}) \cup A'$ finish the limit-point computations?

Part (a): $A = [0, 1)$

Closure: show $[0, 1) \subseteq \overline{A} \subseteq [0, 1]$, then prove $1 \in \overline{A}$ by choosing a point of A in $(1 - r, 1 + r)$:

Isolated points: fix $a \in A$. Given $r > 0$, choose a second point of A inside $(a - r, a + r)$:

Limit points: use the closure/isolated-point relationship to conclude:

Part (b): $A = \mathbb{Z}$

Closure: for $x \notin \mathbb{Z}$, choose $n \in \mathbb{Z}$ with $n < x < n + 1$ and radius $r =$

Isolated points and limit points: choose the ball around $n \in \mathbb{Z}$:

Part (c): $A = \{x \in \mathbb{Q} : 0 < x < 1\}$

Closure: prove 0, points in $(0, 1]$, and no points outside $[0, 1]$ behave correctly:

Isolated points and limit points: use density of $\mathbb{Q}$ to find a second point:

Use this page for parts (a) and (b).

Use this page for part (c) and final cleanup.

Official Solution (Professor)

- (a) Closure: $\overline{A} = [0, 1]$. Since $\overline{A}$ is the smallest closed set containing A , and $[0, 1]$ is one such closed set, we must have

$$[0, 1) \subseteq \overline{A} \subseteq [0, 1].$$

Also $1 \in \overline{A}$: given $r > 0$,

$$(1 - r, 1 + r) \cap [0, 1) \neq \emptyset,$$

since $\max\{1 - r/2, 0\} \in (1 - r, 1 + r) \cap [0, 1)$.

Isolated points: $\emptyset$. Fix $a \in A$. Given $r > 0$, the point

$$\min\left\{a + \frac{r}{2}, \frac{a + 1}{2}\right\}$$

is in $(a - r, a + r) \cap A$ and is not a .

Limit points: $A' = [0, 1]$. By definition, $\overline{A}$ is the union of isolated points and limit points, and there are no isolated points.

- (b) Closure: $\overline{A} = \mathbb{Z}$. We automatically have $\mathbb{Z} \subseteq \overline{A}$. Fix $x \in \mathbb{Z}^c$. Let $n \in \mathbb{Z}$ satisfy $n < x < n + 1$. Then

$$r = \min\{x - n, n + 1 - x\}$$

is positive and $(x - r, x + r) \cap \mathbb{Z} = \emptyset$. Thus $x \notin \overline{A}$.

Isolated points: $\mathbb{Z}$. For $n \in \mathbb{Z}$,

$$(n - 1, n + 1) \cap \mathbb{Z} = \{n\}.$$

Limit points: $A' = \emptyset$. Every point of A is isolated.

- (c) Closure: $\overline{A} = [0, 1]$. First, $A \subseteq [0, 1]$, and $[0, 1]$ is closed, so $\overline{A} \subseteq [0, 1]$. The point 0 is adherent to A : given $r > 0$, choose $n > \max\{1/r, 1\}$, so $1/n \in (-r, r) \cap A$. Now fix $x \in (0, 1]$. By density of $\mathbb{Q}$ in $\mathbb{R}$, for every $r > 0$ there is $q \in \mathbb{Q}$ between $\max\{x - r, 0\}$ and x , and this $q \in (x - r, x + r) \cap A$.

Isolated points: $\emptyset$. Given $x \in A$ and $r > 0$, choose $q \in \mathbb{Q}$ as above. This q lies in $(x - r, x + r) \cap A$ and $q \neq x$.

Limit points: $A' = [0, 1]$. There are no isolated points.

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 2

Problem Statement

Homework 9, Problem 2

Let (X, d) be a metric space, let A be a nonempty subset of X , and let U be an open subset of X .

- (a) Prove that $U \cap \overline{A} \subseteq \overline{U \cap A}$.
- (b) Prove that $\overline{U \cap \overline{A}} = \overline{U \cap A}$. (Hint: Use part (a) and properties of closure from lecture.)
- (c) Conclude that if $U \cap A = \emptyset$, then $U \cap \overline{A} = \emptyset$.

Topic: _____

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

1. Why is the assumption that U is open essential in part (a)?

2. After fixing $x \in U \cap \overline{A}$ and $r > 0$, why do we introduce a second radius $s > 0$?

3. Why is $\tilde{r} = \min\{r, s\}$ the right radius to use?

4. Which closure property turns part (a) into the first inclusion in part (b)?

5. In part (c), why does $\overline{U \cap \overline{A}} = \emptyset$ force $U \cap \overline{A} = \emptyset$?

Part (a): prove $U \cap \overline{A} \subseteq \overline{U \cap A}$.

Fix $x \in \underline{\hspace{2cm}}$. **To show** $x \in \overline{U \cap A}$, **fix** $r > 0$ **and aim to find a point in:**

Use openness of U : since $x \in U$, choose $s > 0$ such that:

Use $x \in \overline{A}$ **with** $\tilde{r} = \min\{r, s\}$ **to choose** $y \in$:

Check the three memberships $y \in A$, $y \in U$, **and** $y \in B_r(x)$:

Part (b): prove equality of closures.

First inclusion from part (a) and monotonicity of closure:

Second inclusion from $A \subseteq \overline{A}$:

Part (c): empty-set conclusion.

Start from $U \cap A = \emptyset$, take closures, use part (b), and then use $B \subseteq \overline{B}$:

Use this page for parts (a) and (b).

Use this page for part (c) and final cleanup.

Official Solution (Professor)

- (a) Fix $x \in U \cap \overline{A}$. We show $x \in \overline{U \cap A}$, i.e., for any $r > 0$, $B_r(x) \cap U \cap A$ is nonempty. Fix $r > 0$. Since $x \in U$ and U is open, there exists $s > 0$ such that $B_s(x) \subseteq U$. Set $\tilde{r} = \min\{r, s\}$. Since $x \in \overline{A}$, the set $B_{\tilde{r}}(x) \cap A$ is nonempty, so choose $y \in B_{\tilde{r}}(x) \cap A$. Then $y \in A$. Also $y \in B_r(x)$, since $d(y, x) < \tilde{r} \leq r$. Finally $y \in U$, since $d(y, x) < \tilde{r} \leq s$, so $y \in B_s(x) \subseteq U$. Hence $y \in B_r(x) \cap U \cap A$, as desired.
- (b) Recall that $B \subseteq C$ implies $\overline{B} \subseteq \overline{C}$. Applying this to part (a),

$$\overline{U \cap \overline{A}} \subseteq \overline{U \cap A}.$$

On the other hand, $A \subseteq \overline{A}$, so $U \cap A \subseteq U \cap \overline{A}$. Therefore

$$\overline{U \cap A} \subseteq \overline{U \cap \overline{A}}.$$

Together,

$$\overline{U \cap \overline{A}} = \overline{U \cap A}.$$

- (c) For any set B , $B \subseteq \overline{B}$. If $U \cap A = \emptyset$, then $\overline{U \cap A} = \emptyset$. By part (b),

$$\overline{U \cap \overline{A}} = \emptyset.$$

Therefore $U \cap \overline{A} = \emptyset$.

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 3

Problem Statement

Homework 9, Problem 3

Recall that d_2 is a metric on $\mathbb{R}^2$. Consider the subspace

$$Y = \{(x, 0) : x \in \mathbb{R}\}$$

with the induced metric d_Y .

- (a) Provide an example of a set $A \subseteq Y$ where A is open in (Y, d_Y) but not open in $(\mathbb{R}^2, d_2)$.
- (b) Prove that for any $B \subseteq Y$, if B is closed in (Y, d_Y) then B is closed in $(\mathbb{R}^2, d_2)$.

Topic: _____

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

1. What does a ball in the subspace Y look like compared with a ball in $\mathbb{R}^2$?

2. Why is the line segment $(-1, 1) \times \{0\}$ open in Y ?

3. Why does the point $(0, r/2)$ prove that this set is not open in $\mathbb{R}^2$?

4. What lecture result lets us reduce part (b) to proving that Y is closed in $\mathbb{R}^2$?

5. How does the vertical coordinate $x_2 \neq 0$ give a ball contained in Y^c ?

Part (a): example open in Y , not open in $\mathbb{R}^2$.

Choose $A =$

To prove A is open in Y , fix $x = (x_1, 0) \in A$ and choose $r =$

If $y \in B_r^Y(x)$, show $|y_1| < 1$, hence $y \in A$:

To prove A is not open in $\mathbb{R}^2$, fix $r > 0$ around $(0, 0)$ and find an off-axis point:

Part (b): closedness transfer.

State the lecture corollary being used:

Part (b): closedness transfer, continued.

Prove Y is closed by fixing $x = (x_1, x_2) \in Y^c$, so $x_2 \neq 0$, and choosing radius $r =$

Show $B_r^{\mathbb{R}^2}(x) \subseteq Y^c$ using the inequality involving $|x_2 - y_2|$:

Use this page for part (a).

Use this page for part (b).

Official Solution (Professor)

(a) Consider

$$A = \{(x_1, 0) : -1 < x_1 < 1\}.$$

First, A is open in (Y, d_Y) . Fix $x = (x_1, 0) \in A$, and set $r = 1 - |x_1|$. Then $r > 0$, and

$$B_r^Y(x) = \{y \in Y : d_Y(x, y) < r\} \subseteq A,$$

since

$$y \in B_r^Y(x) \implies r > d_Y(x, y) = \sqrt{|x_1 - y_1|^2 + |0 - 0|^2} = |x_1 - y_1|$$

and therefore

$$|y_1| \leq |x_1| + |y_1 - x_1| < |x_1| + r = 1.$$

Next, A is not open in $(\mathbb{R}^2, d_2)$, because $(0, 0) \in A$ is not an interior point. For any $r > 0$, the point $(0, r/2)$ lies in $B_r^{\mathbb{R}^2}((0, 0))$ but not in A .

(b) By the corollary from Lecture 24, it suffices to show that Y is closed in $(\mathbb{R}^2, d_2)$. Fix $x = (x_1, x_2) \in Y^c$. Then $x_2 \neq 0$. Set $r = |x_2|$. If $y \in B_r^{\mathbb{R}^2}(x)$, then

$$r > d_2(x, y) = \sqrt{|x_1 - y_1|^2 + |x_2 - y_2|^2} \geq |x_2 - y_2|.$$

Hence

$$|y_2| \geq |x_2| - |x_2 - y_2| > |x_2| - r = 0.$$

Thus $y_2 \neq 0$, so $y \in Y^c$. Therefore $B_r^{\mathbb{R}^2}(x) \subseteq Y^c$. Since every point of Y^c is interior to Y^c , Y^c is open, and Y is closed.

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 4

Problem Statement

Homework 9, Problem 4

Let (X, d) be a metric space, $\{x_n\}_{n \geq 1}$ a sequence in X , and $x \in X$. Prove that the sequence $\{x_n\}_{n \geq 1}$ converges to x if and only if for every open set U that contains x , we have $x_n \in U$ for all but finitely many $n \in \mathbb{N}$.

Topic: _____

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

1. In the forward direction, how does openness of U connect an arbitrary neighborhood to a ball around x ?

2. What does “for all but finitely many n ” mean in terms of choosing an N ?

3. In the reverse direction, why is $B_r(x)$ the correct open set to test?

4. Why is it enough to prove $x_n \in B_r(x)$ eventually for each $r > 0$?

5. What is the main logical difference between the two directions?

Forward direction: assume $x_n \rightarrow x$.

Fix an open set $U \subseteq X$ with $x \in U$. By openness, choose $r > 0$ such that:

By convergence, choose N such that for all $n \geq N$:

Conclude $x_n \in U$ for all but finitely many n :

Reverse direction: assume eventual membership in every open $U \ni x$.

Fix $r > 0$. Apply the hypothesis to the open set:

Convert “all but finitely many” into an N so that for all $n \geq N$:

Translate $x_n \in B_r(x)$ into the metric convergence inequality:

Conclude because $r > 0$ was arbitrary:

Use this page for both directions.

Official Solution (Professor)

$\Rightarrow$ Suppose $x_n \rightarrow x$ in (X, d) as $n \rightarrow \infty$. Fix an open set $U \subseteq X$ such that $x \in U$. Since U is open, there exists $r > 0$ such that $B_r(x) \subseteq U$. By definition of convergence, there exists N such that $d(x_n, x) < r$ for all $n \geq N$. Therefore $x_n \in B_r(x) \subseteq U$ for all $n \geq N$.

$\Leftarrow$ Suppose that for every open set U containing x , we have $x_n \in U$ for all but finitely many $n \in \mathbb{N}$. Fix $r > 0$. Apply the assumption to the open set $B_r(x)$. Then $x_n \in B_r(x)$ for all but finitely many $n \in \mathbb{N}$. Let N be larger than the largest index for which $x_n \notin B_r(x)$. Then for every $n \geq N$, $x_n \in B_r(x)$, and hence $d(x_n, x) < r$. Since $r > 0$ was arbitrary, $\{x_n\}_{n \geq 1}$ converges to x in (X, d) .

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 5

Problem Statement

Homework 9, Problem 5

Recall that d_2 is a metric on $\mathbb{R}^2$. Suppose that $x^n = (x_1^n, x_2^n)$ is a sequence of points in $\mathbb{R}^2$.

- (a) Prove that $|x_1^n - x_1| \leq d_2(x^n, x)$ for any $x = (x_1, x_2) \in \mathbb{R}^2$.
- (b) Prove that the sequence $\{x^n\}_{n \geq 1}$ converges in $(\mathbb{R}^2, d_2)$ if and only if the sequences $\{x_1^n\}_{n \geq 1}$ and $\{x_2^n\}_{n \geq 1}$ of real numbers both converge.

Topic: _____

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

1. Why does the Euclidean distance $d_2(x^n, x)$ control each coordinate difference?

2. In the forward direction, what is squeezed between 0 and $d_2(x^n, x)$?

3. Why do the coordinate sequences converge to the coordinates of the same point x ?

4. In the reverse direction, how do the real limit laws imply $d_2(x^n, x) \rightarrow 0$?

5. What would change if the problem were in $\mathbb{R}^3$ or $\mathbb{R}^m$?

Part (a): coordinate estimate.

Write the formula for $d_2(x^n, x)$:

Use $|x_2^n - x_2|^2 \geq 0$ to get the desired inequality:

Part (b), forward direction.

Assume $x^n \rightarrow x = (x_1, x_2)$ in $(\mathbb{R}^2, d_2)$. Then $d_2(x^n, x) \rightarrow$

Use part (a) and squeeze to prove $x_1^n \rightarrow x_1$:

Repeat the same argument for the second coordinate:

Part (b), reverse direction.

Assume $x_1^n \rightarrow x_1$ and $x_2^n \rightarrow x_2$. Set $x =$

Use the formula for $d_2(x^n, x)$ and the limit laws to conclude:

Use this page for part (a) and the forward direction of part (b).

Use this page for the reverse direction of part (b).

Official Solution (Professor)

(a) Recall that

$$d_2(x^n, x) = \left(|x_1^n - x_1|^2 + |x_2^n - x_2|^2\right)^{1/2}.$$

Since $|x_2^n - x_2|^2 \geq 0$,

$$d_2(x^n, x) \geq \left(|x_1^n - x_1|^2 + 0\right)^{1/2} \geq |x_1^n - x_1|.$$

(b) $\Rightarrow$ Suppose $\{x^n\}_{n \geq 1}$ converges to x in $(\mathbb{R}^2, d_2)$. Then

$$d_2(x^n, x) \rightarrow 0.$$

By part (a),

$$0 \leq |x_1^n - x_1| \leq d_2(x^n, x).$$

By the squeeze theorem, $|x_1^n - x_1| \rightarrow 0$, so $\{x_1^n\}_{n \geq 1}$ converges to x_1 . Similarly,

$$0 \leq |x_2^n - x_2| \leq d_2(x^n, x),$$

so $\{x_2^n\}_{n \geq 1}$ converges to x_2 .

$\Leftarrow$ Suppose $\{x_1^n\}_{n \geq 1}$ converges to x_1 and $\{x_2^n\}_{n \geq 1}$ converges to x_2 . Then

$$|x_1^n - x_1| \rightarrow 0 \quad \text{and} \quad |x_2^n - x_2| \rightarrow 0.$$

By the limit laws,

$$d_2(x^n, x) = \left(|x_1^n - x_1|^2 + |x_2^n - x_2|^2\right)^{1/2} \rightarrow 0.$$

Thus $\{x^n\}_{n \geq 1}$ converges to x in $(\mathbb{R}^2, d_2)$.

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 6

Problem Statement

Homework 9, Problem 6

Let (X, d) be a metric space and $\{x_n\}_{n \geq 1}$ a sequence in X . Prove that if $\{x_n\}_{n \geq 1}$ converges to $x \in X$, then for any $y \in X$, the sequence $\{d(x_n, y)\}_{n \geq 1}$ of real numbers converges to $d(x, y)$.

Topic: _____

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

1. What real-valued sequence are we trying to prove converges?

2. Why should the triangle inequality compare $d(x_n, y)$ and $d(x, y)$?

3. How do two applications of the triangle inequality produce an absolute value estimate?

4. What quantity goes to 0, and why does it squeeze the distance error?

5. How is this problem a sequence version of continuity of the map $z \mapsto d(z, y)$?

Setup.

Fix $y \in X$. Since $x_n \rightarrow x$, we know $d(x_n, x) \rightarrow$

Triangle inequality estimates.

First apply triangle inequality to bound $d(x_n, y)$ in terms of $d(x_n, x)$ and $d(x, y)$:

Rearrange to get $d(x_n, y) - d(x, y) \leq$

Apply triangle inequality again to bound $d(x, y)$ in terms of $d(x, x_n)$ and $d(x_n, y)$:

Rearrange to get $d(x, y) - d(x_n, y) \leq$

Finish.

Combine the two inequalities into an absolute value estimate:

Use squeeze theorem to conclude:

Use this page for the complete proof.

Official Solution (Professor)

Fix $y \in X$. By the triangle inequality,

$$\begin{aligned}d(x_n, y) &\leq d(x_n, x) + d(x, y) &\implies d(x_n, y) - d(x, y) &\leq d(x_n, x), \\d(x, y) &\leq d(x, x_n) + d(x_n, y) &\implies d(x, y) - d(x_n, y) &\leq d(x_n, x).\end{aligned}$$

Together,

$$|d(x_n, y) - d(x, y)| \leq d(x_n, x).$$

Since $\{x_n\}_{n \geq 1}$ converges to x in (X, d) , the right-hand side converges to 0. Therefore, by the squeeze theorem,

$$|d(x_n, y) - d(x, y)| \rightarrow 0,$$

and thus $d(x_n, y)$ converges to $d(x, y)$.

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Homework 10

Problem 1

Problem Statement

Homework 10, Problem 1

Let (X, d) be a metric space and let $Y \subseteq X$. We call the set Y complete if the subspace Y with the induced metric is complete. Prove that if Y is complete, then Y is closed in X . (Hint: It suffices to show $Y \subseteq \bar{Y}$. Use sequences.)

Topic: _____

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

1. What does it mean for Y to be complete as a subspace of X ?

2. What sequential characterization of closure should be used after fixing $x \in \overline{Y}$?

3. Why is a sequence converging in X automatically Cauchy in X ?

4. Why is the same sequence Cauchy in the induced metric on Y ?

5. Where is uniqueness of limits in a metric space used?

Goal: prove Y is closed.

It suffices to show $\overline{Y} \subseteq Y$. Fix:

Use the sequential characterization of closure to get a sequence $\{y_n\} \subseteq Y$ such that:

Explain why $\{y_n\}$ is Cauchy in Y :

Use completeness of Y to get $y \in Y$ with:

Compare convergence in Y with convergence in X , then use uniqueness of limits:

Conclude $\overline{Y} \subseteq Y$, hence Y is closed:

Use this page for the complete proof.

Student-Derived Solution (No official solution available)

We show that $\overline{Y} \subseteq Y$. Since always $Y \subseteq \overline{Y}$, this implies $\overline{Y} = Y$, and hence Y is closed in X .

Let $x \in \overline{Y}$. By the sequential characterization of closure, there exists a sequence $\{y_n\}_{n \geq 1}$ in Y such that

$$y_n \rightarrow x \quad \text{in } (X, d).$$

Every convergent sequence in a metric space is Cauchy, so $\{y_n\}$ is Cauchy in (X, d) . Since the induced metric on Y is the restriction of d , the same sequence is Cauchy in (Y, d_Y) .

Because Y is complete, there is some $y \in Y$ such that

$$y_n \rightarrow y \quad \text{in } (Y, d_Y).$$

Convergence in the subspace metric is the same distance condition as convergence in X , so $y_n \rightarrow y$ in (X, d) . Thus the sequence $\{y_n\}$ converges to both x and y in X . By uniqueness of limits in metric spaces, $x = y$. Since $y \in Y$, we have $x \in Y$. Therefore $\overline{Y} \subseteq Y$, so Y is closed in X .

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 2

Problem Statement

Homework 10, Problem 2

Let (X, d) be a metric space. Prove that if $A, B \subseteq X$ are both compact then $A \cup B$ is compact.

Topic: _____

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

1. What definition of compactness is most direct for a finite union?

2. Why does an open cover of $A \cup B$ automatically cover A and B separately?

3. What finite subcover do you get from compactness of A ?

4. What finite subcover do you get from compactness of B ?

5. Why is the union of the two finite index sets still finite and enough to cover $A \cup B$?

Open-cover setup.

Let $\{U_i\}_{i \in I}$ be an arbitrary open cover of $A \cup B$:

Restrict the same cover to A , and use compactness of A to choose a finite $J_A \subseteq I$:

Restrict the same cover to B , and use compactness of B to choose a finite $J_B \subseteq I$:

Set $J =$

Check J is finite and $\{U_i\}_{i \in J}$ covers $A \cup B$:

Conclude compactness because the original open cover was arbitrary:

Use this page for the complete proof.

Student-Derived Solution (No official solution available)

Let $\{U_i\}_{i \in I}$ be an open cover of $A \cup B$. Thus each U_i is open in X , and

$$A \cup B \subseteq \bigcup_{i \in I} U_i.$$

Since $A \subseteq A \cup B$, the same family is an open cover of A . Because A is compact, there is a finite set $J_A \subseteq I$ such that

$$A \subseteq \bigcup_{i \in J_A} U_i.$$

Similarly, $\{U_i\}_{i \in I}$ is an open cover of B , so compactness of B gives a finite set $J_B \subseteq I$ such that

$$B \subseteq \bigcup_{i \in J_B} U_i.$$

Set $J = J_A \cup J_B$. Then J is finite, and

$$A \cup B \subseteq \left(\bigcup_{i \in J_A} U_i \right) \cup \left(\bigcup_{i \in J_B} U_i \right) = \bigcup_{i \in J} U_i.$$

Thus every open cover of $A \cup B$ has a finite subcover. Therefore $A \cup B$ is compact.

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 3

Problem Statement

Homework 10, Problem 3

Let (X, d) be a metric space.

- (a) Prove that if $A \subseteq B \subseteq X$, A is closed in X , and B is compact, then A is compact. (Hint: Given an open cover of A , we can add A^c to get an open cover of B .)
- (b) Prove that if $C, D \subseteq X$ are both compact, then $C \cap D$ is compact. (Hint: Use (a).)

Topic: _____

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

1. Why does adding A^c to an open cover of A help cover all of B ?

2. Where is the assumption that A is closed used?

3. After compactness of B gives a finite subcover, why can A^c be discarded when covering A ?

4. Which lecture result says compact subsets of a metric space are closed?

5. In part (b), which compact set plays the role of B in part (a)?

Part (a): closed subset of a compact set.

Let $\{U_i\}_{i \in I}$ be an open cover of A . Since A is closed, A^c is:

Show $\{A^c\} \cup \{U_i\}_{i \in I}$ covers B :

Use compactness of B to extract a finite subcover:

Restrict back to A and remove the A^c part:

Part (b): compact intersection.

Use compactness $\Rightarrow$ closed to show $C \cap D$ is closed in X :

Apply part (a) with $A = C \cap D$ and $B =$

Use this page for part (a).

Use this page for part (b).

Official Solution (Professor)

(a) Let $\{U_i\}_{i \in I}$ be an open cover of A . Since A is closed, A^c is open. Thus

$$\{A^c\} \cup \{U_i\}_{i \in I}$$

is an open cover of B , because

$$B \subseteq X = A^c \cup A \subseteq A^c \cup \bigcup_{i \in I} U_i.$$

Since B is compact, there is a finite subcover for B . We may write it as

$$\{A^c\} \cup \{U_i\}_{i \in J}$$

for some finite $J \subseteq I$. Then

$$B \subseteq A^c \cup \bigcup_{i \in J} U_i.$$

Since $A \subseteq B$ and $A \cap A^c = \emptyset$, it follows that

$$A \subseteq \bigcup_{i \in J} U_i.$$

Hence $\{U_i\}_{i \in J}$ is a finite subcover of A . Therefore A is compact.

(b) Since C and D are compact, they are closed by Lecture 27. Hence $C \cap D$ is closed. Also $C \cap D \subseteq C$, and C is compact. By part (a), $C \cap D$ is compact.

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 4

Problem Statement

Homework 10, Problem 4

Let (X, d) be a metric space and let $Y \subseteq X$. Prove that if Y is compact, then Y is complete.

Topic: _____

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

1. To prove completeness, what kind of sequence in Y should you start with?

2. Which compactness theorem gives a convergent subsequence in Y ?

3. Why is a convergent subsequence not enough by itself to prove completeness?

4. How does the Cauchy property transfer convergence from a subsequence to the full sequence?

5. Why does the final limit lie in Y ?

Setup: prove every Cauchy sequence in Y converges in Y .

Let $\{y_n\}$ be a Cauchy sequence in Y :

Use compactness and Lecture 26 to obtain a subsequence $\{y_{k_n}\}$ and point $y \in Y$ such that:

Fix $r > 0$. From subsequence convergence choose N_1 such that:

From the Cauchy property choose N_2 such that:

Use triangle inequality with y_n , y_{k_n} , and y :

Conclude the whole sequence converges to $y \in Y$:

Use this page for the complete proof.

Official Solution (Professor)

Let $\{y_n\}_{n \geq 1}$ be a Cauchy sequence in Y . We show that it converges to some point of Y .

Since Y is compact, by Lecture 26 there exists a subsequence $\{y_{k_n}\}_{n \geq 1}$ that converges to some point $y \in Y$.

We prove the whole sequence converges to y . Fix $r > 0$. Since $y_{k_n} \rightarrow y$, there exists N_1 such that

$$d(y_{k_n}, y) < \frac{r}{2} \quad \text{for all } n \geq N_1.$$

Since $\{y_n\}$ is Cauchy, there exists N_2 such that

$$d(y_n, y_m) < \frac{r}{2} \quad \text{for all } n, m \geq N_2.$$

Set $N = \max\{N_1, N_2\}$. For $n \geq N$, we have $k_n \geq n \geq N$, and so

$$d(y_n, y) \leq d(y_n, y_{k_n}) + d(y_{k_n}, y) < \frac{r}{2} + \frac{r}{2} = r.$$

Since $r > 0$ was arbitrary, $y_n \rightarrow y$. Because $y \in Y$, every Cauchy sequence in Y converges in Y . Therefore Y is complete.

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Homework 11

Problem 1

Problem Statement

Homework 11, Problem 1

Consider the metric space $(\mathbb{R}, d)$ with the usual metric $d(x, y) = |x - y|$. Note that the set $\mathbb{Q}$ is not an interval, and so from lecture we know that it must be disconnected.

- (a) Find two sets $A, B \subseteq \mathbb{R}$ that are nonempty and separated so that $\mathbb{Q} = A \cup B$, and prove your answer.
- (b) Find a set $\emptyset \neq C \subsetneq \mathbb{Q}$ that is both open and closed in $\mathbb{Q}$, and prove your answer.

Topic: _____

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

1. Why is cutting $\mathbb{Q}$ at an irrational number better than cutting it at a rational number?

2. What does it mean for two subsets of $\mathbb{R}$ to be separated?

3. What are the closures in $\mathbb{R}$ of $(-\infty, \sqrt{2}) \cap \mathbb{Q}$ and $(\sqrt{2}, \infty) \cap \mathbb{Q}$?

4. Why does $\sqrt{2} \notin \mathbb{Q}$ make the separation work?

5. How does the subspace characterization of open and closed sets in $\mathbb{Q}$ prove part (b)?

Part (a): separated decomposition of $\mathbb{Q}$.
Choose an irrational cut point and define $A =$

Define $B =$

Check A and B are nonempty and $A \cup B = \mathbb{Q}$:

Compute $\overline{A}$ and $\overline{B}$ in $\mathbb{R}$:

Use those closures to prove $\overline{A} \cap B = \emptyset$ and $A \cap \overline{B} = \emptyset$:

Part (b): clopen subset of $\mathbb{Q}$.

Choose $C =$

Prove C is nonempty and proper:

Show C is open in $\mathbb{Q}$ and closed in $\mathbb{Q}$ using subspace results:

Use this page for part (a).

Use this page for part (b).

Student-Derived Solution (No official solution available)

(a) Define

$$A = (-\infty, \sqrt{2}) \cap \mathbb{Q}, \quad B = (\sqrt{2}, \infty) \cap \mathbb{Q}.$$

Both sets are nonempty, for instance $1 \in A$ and $2 \in B$. Also $A \cup B = \mathbb{Q}$, since every rational number is either less than $\sqrt{2}$ or greater than $\sqrt{2}$, and $\sqrt{2} \notin \mathbb{Q}$. In $\mathbb{R}$,

$$\overline{A} = (-\infty, \sqrt{2}], \quad \overline{B} = [\sqrt{2}, \infty).$$

Hence

$$\overline{A} \cap B = (-\infty, \sqrt{2}] \cap ((\sqrt{2}, \infty) \cap \mathbb{Q}) = \emptyset,$$

and similarly

$$A \cap \overline{B} = ((-\infty, \sqrt{2}) \cap \mathbb{Q}) \cap [\sqrt{2}, \infty) = \emptyset.$$

Therefore A and B are separated, nonempty, and satisfy $\mathbb{Q} = A \cup B$.

(b) Let

$$C = (-\infty, \sqrt{2}) \cap \mathbb{Q}.$$

Then $C \neq \emptyset$, since $1 \in C$, and $C \subsetneq \mathbb{Q}$, since $2 \in \mathbb{Q} \setminus C$.

The set $(-\infty, \sqrt{2})$ is open in $\mathbb{R}$, so $C = (-\infty, \sqrt{2}) \cap \mathbb{Q}$ is open in the subspace $\mathbb{Q}$. Also $(-\infty, \sqrt{2}]$ is closed in $\mathbb{R}$, and since $\sqrt{2} \notin \mathbb{Q}$,

$$(-\infty, \sqrt{2}] \cap \mathbb{Q} = (-\infty, \sqrt{2}) \cap \mathbb{Q} = C.$$

Thus C is closed in $\mathbb{Q}$. Therefore C is a nonempty proper subset of $\mathbb{Q}$ that is both open and closed in $\mathbb{Q}$.

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 2

Problem Statement

Homework 11, Problem 2

Let (X, d) be a metric space.

- (a) Prove that if $F \subseteq X$ is disconnected and closed in X , then there exist $A, B \subseteq X$ that are nonempty, separated, and closed in X so that $F = A \cup B$.
- (b) Prove that if $U \subseteq X$ is disconnected and open in X , then there exist $A, B \subseteq X$ that are nonempty, separated, and open in X so that $U = A \cup B$.

Topic: _____

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

1. Which Lecture 29 characterization of disconnectedness gives a nontrivial clopen subset of F or U ?

2. In part (a), why does closed in F plus F closed in X imply closed in X ?

3. In part (b), why does open in U plus U open in X imply open in X ?

4. Why are the two pieces nonempty?

5. How do closedness or openness of the pieces imply they are separated?

Part (a): closed disconnected set.

Use disconnectedness of F to choose $A \neq \emptyset, F$ that is both open and closed in F . Set $B =$

Explain why A and B are nonempty and $F = A \cup B$:

Use F closed in X to prove A and B are closed in X :

Use closedness and disjointness to prove separatedness:

Part (b): open disconnected set.

Use disconnectedness of U to choose $A \neq \emptyset, U$ clopen in U , and set $B =$

Use U open in X to prove A and B are open in X :

Use complements of open sets to prove separatedness:

Use this page for part (a).

Use this page for part (b).

Official Solution (Professor)

- (a) Since F is disconnected, by Lecture 29 there exists a set $A \neq \emptyset, F$ that is both open and closed in F . Set $B = F \setminus A$.

The sets A and B are nonempty because $A \neq \emptyset$ and $A \neq F$. Since A is closed in F and F is closed in X , A is closed in X by Lecture 24. Also $B = F \setminus A$ is closed in F , because A is open in F ; hence B is closed in X .

Finally, $A \cap B = \emptyset$. Since A and B are closed in X , $\overline{A} = A$ and $\overline{B} = B$. Thus

$$\overline{A} \cap B = \emptyset, \quad A \cap \overline{B} = \emptyset.$$

Therefore A and B are nonempty, separated, closed in X , and $F = A \cup B$.

- (b) Since U is disconnected, by Lecture 29 there exists a set $A \neq \emptyset, U$ that is both open and closed in U . Set $B = U \setminus A$.

The sets A and B are nonempty. Since A is open in U and U is open in X , A is open in X by Lecture 24. Since A is closed in U , $B = U \setminus A$ is open in U ; because U is open in X , B is open in X .

Since $A \cap B = \emptyset$, we have $A \subseteq B^c$. Because B is open, B^c is closed, so $\overline{A} \subseteq B^c$, and hence $\overline{A} \cap B = \emptyset$. Similarly, $\overline{B} \cap A = \emptyset$. Therefore A and B are separated, open in X , nonempty, and $U = A \cup B$.

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 3

Problem Statement

Homework 11, Problem 3

Suppose that (X, d) is a metric space such that for any pair of points $a, b \in X$, there exists a connected set $C \subseteq X$ so that $a, b \in C$. Prove that X is connected.

Topic: _____

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

1. Why is contradiction the natural proof method here?

2. What does a separation of X give you?

3. Why can you choose one point a from one side and one point b from the other?

4. Which Lecture 29 result says a connected set inside a union of separated sets lies in one side?

5. Why does the connected set containing both a and b contradict the separation?

Contradiction setup.

Assume X is disconnected. Then choose nonempty separated sets $A, B \subseteq X$ such that:

Choose $a \in A$ and $b \in B$. By hypothesis, choose connected $C \subseteq X$ such that:

Since $C \subseteq X = A \cup B$, apply the connected-set/separated-union result:

Use $a \in A \cap C$ and $b \in B \cap C$ to rule out both alternatives:

Conclude X is connected:

Use this page for the complete proof.

Official Solution (Professor)

Suppose toward a contradiction that X is disconnected. Then there exist sets $A, B \subseteq X$ such that

$$X = A \cup B, \quad A \neq \emptyset, \quad B \neq \emptyset,$$

and A and B are separated.

Choose $a \in A$ and $b \in B$. By the hypothesis, there exists a connected set $C \subseteq X$ such that $a, b \in C$. Since $C \subseteq X = A \cup B$, and A and B are separated, Lecture 29 implies that either $C \subseteq A$ or $C \subseteq B$.

Because $a \in A \cap C$, the only possible case is $C \subseteq A$. But then $B \cap C = \emptyset$, contradicting $b \in B \cap C$. Therefore X cannot be disconnected, so X is connected.

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 4

Problem Statement

Homework 11, Problem 4

Let (X, d) be a metric space and $A \subseteq X$. Prove that if A is connected, then $\overline{A}$ is connected.

Topic: _____

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

1. Why should the proof begin by assuming $\overline{A}$ is disconnected?

2. If $\overline{A} = B \cup C$ with B, C separated, what does connectedness of $A \subseteq B \cup C$ imply?

3. Why can one assume $A \subseteq B$ after relabeling?

4. How does $A \subseteq B$ imply $\overline{A} \subseteq \overline{B}$?

5. Why does separatedness then force the other side to be empty?

Contradiction setup.

Assume $\overline{A}$ is disconnected, so $\overline{A} = B \cup C$ with B, C nonempty and separated:

Apply connectedness of A to the separated union $B \cup C$:

After relabeling, assume $A \subseteq B$. Then conclude $\overline{A} \subseteq$

Use separatedness $\overline{B} \cap C = \emptyset$ to show $\overline{A} \cap C = \emptyset$:

Explain why this contradicts $C \neq \emptyset$ and $C \subseteq \overline{A}$:

Use this page for the complete proof.

Official Solution (Professor)

Suppose toward a contradiction that $\overline{A}$ is disconnected. Then there exist sets $B, C \subseteq X$ such that

$$\overline{A} = B \cup C, \quad B \neq \emptyset, \quad C \neq \emptyset,$$

and B and C are separated.

Since A is connected and $A \subseteq \overline{A} = B \cup C$, Lecture 29 implies that either $A \subseteq B$ or $A \subseteq C$. After swapping B and C if necessary, assume $A \subseteq B$. Then

$$\overline{A} \subseteq \overline{B}.$$

Because B and C are separated, $\overline{B} \cap C = \emptyset$. Hence

$$\overline{A} \cap C = \emptyset.$$

But $C \subseteq \overline{A}$, since $\overline{A} = B \cup C$. Therefore $C = \emptyset$, a contradiction. Thus $\overline{A}$ is connected.

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 5

Problem Statement

Homework 11, Problem 5

Let (X, d) be a metric space and $A, B \subseteq X$. Prove that if A and B are both connected and $A \cap B \neq \emptyset$, then $A \cup B$ is connected.

Topic: _____

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

1. Why does the assumption $A \cap B \neq \emptyset$ prevent A and B from lying in different sides of a separation?

2. If $A \cup B = C \cup D$ is a separation, what does connectedness of A imply?

3. What does connectedness of B imply?

4. How does a shared point force A and B into the same separated side?

5. Why does this make the other side empty?

Contradiction setup.

Assume $A \cup B$ is disconnected. Choose nonempty separated $C, D \subseteq X$ such that:

Use connectedness of $A \subseteq C \cup D$ to conclude:

After relabeling, assume $A \subseteq C$. Use $A \cap B \neq \emptyset$ to show B has a point in C :

Use connectedness of B to conclude $B \subseteq C$:

Then $A \cup B \subseteq C$, contradicting:

Use this page for the complete proof.

Official Solution (Professor)

Suppose toward a contradiction that $A \cup B$ is disconnected. Then there exist sets $C, D \subseteq X$ such that

$$A \cup B = C \cup D, \quad C, D \neq \emptyset,$$

and C and D are separated.

Since $A \subseteq C \cup D$ and A is connected, Lecture 29 implies that either $A \subseteq C$ or $A \subseteq D$. After swapping C and D , assume $A \subseteq C$.

Similarly, since $B \subseteq C \cup D$ and B is connected, either $B \subseteq C$ or $B \subseteq D$. But $A \subseteq C$ and $A \cap B \neq \emptyset$, so at least one point of B lies in C . Therefore B cannot be contained in D , and we must have $B \subseteq C$.

Thus $A \cup B \subseteq C$. Since $A \cup B = C \cup D$, this forces $D = \emptyset$, contradicting the separation. Therefore $A \cup B$ is connected.

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 6

Problem Statement

Homework 11, Problem 6

Let (X, d) be a metric space.

- (a) Let $A, B, C \subseteq X$. Prove that if the sets A and B are separated and the sets A and C are separated, then the sets A and $B \cup C$ are separated.
- (b) Prove that if $F, G \subseteq X$ are closed and $F \cup G$ and $F \cap G$ are connected, then F is connected.

Topic: _____

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

1. What two closure-intersection conditions define separated sets?

2. In part (a), how do unions distribute over intersections with $\overline{A}$?

3. For part (b), why is contradiction a natural strategy?

4. If $F = A \cup B$ is a separation, how does connectedness of $F \cap G$ force $F \cap G$ into one side?

5. Why does proving B is separated from $A \cup G$ contradict connectedness of $F \cup G$?

Part (a): separated from a union.

Write what it means that A is separated from B and from C :

Compute $\overline{A} \cap (B \cup C)$:

Use $\overline{B \cup C} = \overline{B} \cap \overline{C}$ to compute $A \cap \overline{B \cup C}$:

Conclude A and $B \cup C$ are separated:

Part (b): contradiction.

Assume F is disconnected and choose $F = A \cup B$ with A, B nonempty separated:

Use connectedness of $F \cap G$ to assume, after relabeling, $F \cap G \subseteq A$:

Show B and G are separated, using closedness of G :

Apply part (a) to show B and $A \cup G$ are separated, yielding a separation of $F \cup G$:

Use this page for part (a).

Use this page for part (b).

Official Solution (Professor)

(a) Since A and B are separated,

$$\overline{A} \cap B = \emptyset \quad \text{and} \quad A \cap \overline{B} = \emptyset.$$

Since A and C are separated,

$$\overline{A} \cap C = \emptyset \quad \text{and} \quad A \cap \overline{C} = \emptyset.$$

Therefore

$$\overline{A} \cap (B \cup C) = (\overline{A} \cap B) \cup (\overline{A} \cap C) = \emptyset.$$

Also, by Lecture 23,

$$\overline{B \cup C} = \overline{B} \cup \overline{C}.$$

Hence

$$A \cap \overline{B \cup C} = (A \cap \overline{B}) \cup (A \cap \overline{C}) = \emptyset.$$

Thus A and $B \cup C$ are separated.

(b) Suppose toward a contradiction that F is disconnected. Then there exist $A, B \subseteq X$ such that

$$F = A \cup B, \quad A, B \neq \emptyset,$$

and A and B are separated.

Since $F \cap G$ is connected and $F \cap G \subseteq A \cup B$, Lecture 29 implies that either $F \cap G \subseteq A$ or $F \cap G \subseteq B$. After swapping A and B if necessary, assume $F \cap G \subseteq A$. We claim that B and G are separated. First $B \cap G = \emptyset$, because any point in $B \cap G$ would lie in $F \cap G \subseteq A$, contradicting $A \cap B = \emptyset$. Since G is closed, $\overline{G} = G$, and this gives $B \cap \overline{G} = \emptyset$. Also, if $x \in G = \overline{G}$, then either $x \in G \setminus F$ or $x \in F \cap G \subseteq A$, so $x \notin B$. Hence $\overline{B} \cap G = \emptyset$. Thus B and G are separated.

Since A and B are separated, part (a) implies that B and $A \cup G$ are separated. Both are nonempty, and

$$F \cup G = (A \cup B) \cup G = (A \cup G) \cup B.$$

Therefore $F \cup G$ is disconnected, contradicting the hypothesis. Hence F is connected.

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Homework 12

Problem 1

Problem Statement

Homework 12, Problem 1

Let (X, d_X) and (Y, d_Y) be metric spaces, $f : X \rightarrow Y$, and $x_0 \in X$. Prove that f is continuous at x_0 if and only if for any open set $V \subseteq Y$ containing $f(x_0)$, there is an open set $U \subseteq X$ containing x_0 so that $x \in U$ implies $f(x) \in V$.

Topic: _____

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

1. What is the ε - δ definition of continuity at x_0 ?

2. In the forward direction, how does openness of V produce a ball around $f(x_0)$?

3. Why is $U = B_\delta(x_0)$ the natural choice after applying continuity?

4. In the reverse direction, why should $V = B_\varepsilon(f(x_0))$ be used?

5. How does openness of U produce the δ needed for continuity?

Forward direction.

Assume f is continuous at x_0 . Fix open $V \subseteq Y$ with $f(x_0) \in V$. Choose $\varepsilon > 0$ so that:

Use continuity at x_0 to choose $\delta > 0$ so that:

Set $U =$

Check U is open, contains x_0 , and $x \in U \Rightarrow f(x) \in V$:

Reverse direction.

Assume the open-neighborhood condition. Fix $\varepsilon > 0$ and set $V =$

Use the hypothesis to get open $U \ni x_0$ with $f(U) \subseteq V$:

Use openness of U to choose $\delta > 0$ with $B_\delta(x_0) \subseteq U$:

Conclude the ε - δ condition:

Use this page for the forward direction.

Use this page for the reverse direction.

Student-Derived Solution (No official solution available yet)

$\Rightarrow$ Assume f is continuous at x_0 . Let $V \subseteq Y$ be open with $f(x_0) \in V$. Since V is open, there is $\varepsilon > 0$ such that

$$B_\varepsilon(f(x_0)) \subseteq V.$$

By continuity at x_0 , there is $\delta > 0$ such that

$$d_X(x, x_0) < \delta \implies d_Y(f(x), f(x_0)) < \varepsilon.$$

Let $U = B_\delta(x_0)$. Then U is open, $x_0 \in U$, and if $x \in U$, then $f(x) \in B_\varepsilon(f(x_0)) \subseteq V$.

$\Leftarrow$ Assume the open-set condition. Let $\varepsilon > 0$, and set $V = B_\varepsilon(f(x_0))$. Then V is open and contains $f(x_0)$, so there is an open set $U \subseteq X$ with $x_0 \in U$ and $f(U) \subseteq V$. Since U is open and $x_0 \in U$, there is $\delta > 0$ such that $B_\delta(x_0) \subseteq U$. Thus if $d_X(x, x_0) < \delta$, then $x \in U$, so $f(x) \in V$, meaning

$$d_Y(f(x), f(x_0)) < \varepsilon.$$

Therefore f is continuous at x_0 .

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 2

Problem Statement

Homework 12, Problem 2

Let (X, d_X) and (Y, d_Y) be metric spaces and $f : X \rightarrow Y$ a function. Prove that f is continuous if and only if $f(\overline{A}) \subseteq \overline{f(A)}$ for any $A \subseteq X$.

Topic: _____

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

1. What does $x \in \overline{A}$ mean sequentially in a metric space?

2. In the forward direction, how does continuity move a convergent sequence from X to Y ?

3. Why does $f(a_n) \rightarrow f(x)$ prove $f(x) \in \overline{f(A)}$?

4. In the reverse direction, why choose $A = X \setminus f^{-1}(V)$ for an open $V \subseteq Y$?

5. Why does $x \notin \overline{A}$ produce an open neighborhood U of x with $f(U) \subseteq V$?

Forward direction.

Assume f is continuous. Fix $A \subseteq X$ and $x \in \overline{A}$. Choose a sequence $a_n \in A$ such that:

Use continuity to show $f(a_n) \rightarrow$

Conclude $f(x) \in \overline{f(A)}$, hence $f(\overline{A}) \subseteq \overline{f(A)}$:

Reverse direction.

Assume $f(\overline{A}) \subseteq \overline{f(A)}$ for every $A \subseteq X$. Fix open $V \subseteq Y$ with $f(x) \in V$:

Define $A = X \setminus f^{-1}(V)$. Suppose for contradiction that $x \in \overline{A}$:

Use the closure hypothesis to force $f(x) \in \overline{f(A)}$:

Use $f(A) \subseteq Y \setminus V$ and $Y \setminus V$ closed to contradict $f(x) \in V$:

From $x \notin \overline{A}$, build an open $U \ni x$ with $U \subseteq f^{-1}(V)$, then apply Problem 1:

Use this page for the forward direction.

Use this page for the reverse direction.

Student-Derived Solution (No official solution available yet)

$\Rightarrow$ Assume f is continuous. Let $A \subseteq X$, and let $x \in \overline{A}$. By the sequential characterization of closure, there is a sequence $\{a_n\} \subseteq A$ such that $a_n \rightarrow x$. Since f is continuous,

$$f(a_n) \rightarrow f(x).$$

Each $f(a_n) \in f(A)$, so $f(x) \in \overline{f(A)}$. Therefore

$$f(\overline{A}) \subseteq \overline{f(A)}.$$

$\Leftarrow$ Assume that for every $A \subseteq X$,

$$f(\overline{A}) \subseteq \overline{f(A)}.$$

We prove continuity using Problem 1. Let $V \subseteq Y$ be open and suppose $f(x) \in V$. Set

$$A = X \setminus f^{-1}(V).$$

We claim $x \notin \overline{A}$. If $x \in \overline{A}$, then by the hypothesis

$$f(x) \in f(\overline{A}) \subseteq \overline{f(A)}.$$

But if $a \in A$, then $f(a) \notin V$, so $f(A) \subseteq Y \setminus V$. Since $Y \setminus V$ is closed,

$$\overline{f(A)} \subseteq Y \setminus V.$$

This would imply $f(x) \notin V$, contradicting $f(x) \in V$. Thus $x \notin \overline{A}$.

Since $x \notin \overline{A}$, there exists an open set $U \subseteq X$ containing x such that $U \cap A = \emptyset$. Hence

$$U \subseteq X \setminus A = f^{-1}(V),$$

so $f(U) \subseteq V$. By Problem 1, f is continuous at x . Since $x \in X$ was arbitrary, f is continuous.

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 3

Problem Statement

Homework 12, Problem 3

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the function

$$f(x) = \frac{1}{1 + x^2}.$$

- (a) Prove that f is continuous.
- (b) Find an open set $U \subseteq \mathbb{R}$ so that $f(U)$ is not open.
- (c) Find a closed set $F \subseteq \mathbb{R}$ so that $f(F)$ is not closed.
- (d) Find a set $A \subseteq \mathbb{R}$ so that $f(\overline{A}) \neq \overline{f(A)}$.

Topic: _____

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

1. Which standard limit laws prove continuity of $f(x) = 1/(1 + x^2)$?

2. What is the range of f on all of $\mathbb{R}$?

3. Why is $(0, 1]$ not open in $\mathbb{R}$?

4. Why is $(0, 1]$ not closed in $\mathbb{R}$?

5. For $A = (0, \infty)$, compare $f(\overline{A})$ with $\overline{f(A)}$. Where does the extra endpoint come from?

Part (a): continuity.

Use sequences or limit laws: if $x_n \rightarrow x$, then $x_n^2 \rightarrow$

Conclude $1 + x_n^2 \rightarrow 1 + x^2$, and since denominators stay nonzero:

Parts (b) and (c): image counterexamples.

For an open set, try $U = \mathbb{R}$. Compute $f(U) =$

Explain why that image is not open:

For a closed set, try $F = [0, \infty)$. Compute $f(F) =$

Explain why that image is not closed:

Part (d): closure/image mismatch.

Try $A = (0, \infty)$. Then $\overline{A} =$

Compute $f(\overline{A})$, $f(A)$, and $\overline{f(A)}$:

Use this page for parts (a) and (b).

Use this page for parts (c) and (d).

Student-Derived Solution (No official solution available yet)

(a) Let $x_n \rightarrow x$ in $\mathbb{R}$. Then by the standard limit laws,

$$x_n^2 \rightarrow x^2, \quad 1 + x_n^2 \rightarrow 1 + x^2.$$

Since $1 + x^2 \neq 0$, the reciprocal limit law gives

$$\frac{1}{1 + x_n^2} \rightarrow \frac{1}{1 + x^2}.$$

Hence $f(x_n) \rightarrow f(x)$, so f is continuous.

(b) Let $U = \mathbb{R}$, which is open. Then

$$f(U) = (0, 1].$$

This is not open in $\mathbb{R}$, since $1 \in (0, 1]$ but no open interval around 1 is contained in $(0, 1]$.

(c) Let $F = [0, \infty)$, which is closed in $\mathbb{R}$. Then

$$f(F) = (0, 1].$$

This set is not closed in $\mathbb{R}$, since 0 is a limit point of $(0, 1]$ but $0 \notin (0, 1]$.

(d) Let $A = (0, \infty)$. Then $\overline{A} = [0, \infty)$, so

$$f(\overline{A}) = f([0, \infty)) = (0, 1].$$

On the other hand,

$$f(A) = f((0, \infty)) = (0, 1), \quad \overline{f(A)} = [0, 1].$$

Therefore

$$f(\overline{A}) = (0, 1] \neq [0, 1] = \overline{f(A)}.$$

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 4

Problem Statement

Homework 12, Problem 4

Consider the metric spaces $(\mathbb{R}, |\cdot|)$ and $(\mathbb{R}^2, d_2)$.

- (a) Prove that the functions $f, g : \mathbb{R}^2 \rightarrow \mathbb{R}$ given by $f(x, y) = x$ and $g(x, y) = y$ are continuous. (Hint: Use sequences.)
- (b) Prove that if $h : [0, 1] \rightarrow \mathbb{R}^2$ is a continuous function such that $h(0) = (x_0, y_0)$, $h(1) = (x_1, y_1)$, and $x_0 < 0 < x_1$, then there exists a point $t \in (0, 1)$ so that $h(t)$ lies on the y -axis.

Topic: _____

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

1. How does convergence in $(\mathbb{R}^2, d_2)$ imply coordinatewise convergence?

2. Why does $|x_n - x| \leq d_2((x_n, y_n), (x, y))$ prove continuity of the first projection?

3. What scalar function should be formed from the path h in part (b)?

4. Why is $\phi(t) = f(h(t))$ continuous?

5. How does the Intermediate Value Theorem produce a point on the y -axis?

Part (a): projections are continuous.

Let $(x_n, y_n) \rightarrow (x, y)$ in $(\mathbb{R}^2, d_2)$. Write the coordinate inequality for x_n :

Use squeeze to prove $x_n \rightarrow x$, hence $f(x_n, y_n) \rightarrow f(x, y)$:

Repeat the coordinate inequality for y_n :

Part (b): path crosses the y -axis.

Define $\phi : [0, 1] \rightarrow \mathbb{R}$ by $\phi(t) =$

Explain why ϕ is continuous:

Compute $\phi(0)$ and $\phi(1)$, and record their signs:

Apply the Intermediate Value Theorem to get $t \in (0, 1)$ with $\phi(t) = 0$:

Translate $\phi(t) = 0$ into " $h(t)$ lies on the y -axis":

Use this page for part (a).

Use this page for part (b).

Student-Derived Solution (No official solution available yet)

(a) We prove continuity of $f(x, y) = x$. Let $(x_n, y_n) \rightarrow (x, y)$ in $(\mathbb{R}^2, d_2)$. Then

$$d_2((x_n, y_n), (x, y)) = \sqrt{(x_n - x)^2 + (y_n - y)^2} \rightarrow 0.$$

Since

$$|x_n - x| \leq d_2((x_n, y_n), (x, y)),$$

the squeeze theorem gives $x_n \rightarrow x$. Thus

$$f(x_n, y_n) = x_n \rightarrow x = f(x, y),$$

so f is continuous. The proof for $g(x, y) = y$ is the same: use

$$|y_n - y| \leq d_2((x_n, y_n), (x, y))$$

to get $y_n \rightarrow y$, hence $g(x_n, y_n) \rightarrow g(x, y)$.

(b) Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be the first-coordinate projection from part (a), and define

$$\phi : [0, 1] \rightarrow \mathbb{R}, \quad \phi(t) = f(h(t)).$$

Since h is continuous and f is continuous, $\phi = f \circ h$ is continuous. Also

$$\phi(0) = f(h(0)) = f(x_0, y_0) = x_0 < 0,$$

and

$$\phi(1) = f(h(1)) = f(x_1, y_1) = x_1 > 0.$$

By the Intermediate Value Theorem, there exists $t \in (0, 1)$ such that $\phi(t) = 0$. This means the first coordinate of $h(t)$ is 0, so $h(t) \in \{(0, y) : y \in \mathbb{R}\}$. Therefore $h(t)$ lies on the y -axis.

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 5

Problem Statement

Homework 12, Problem 5

Going forward, you may assume (no proof necessary) any fact about $\sin x$ and $\cos x$ from calculus, such as $\sin, \cos : \mathbb{R} \rightarrow [-1, 1]$ are continuous functions.

Prove that the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ given by

$$f(x, y) = x^2 + y^2 + \sin(x - y)$$

has a global minimum. (Hint: Notice that we cannot directly apply the extreme value theorem here. However, we can apply it to any compact subset of $\mathbb{R}^2$.)

Topic: _____

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

1. Why can the extreme value theorem not be applied directly on all of $\mathbb{R}^2$?

2. Why is f continuous?

3. What lower bound follows from $\sin(x - y) \geq -1$?

4. Why does the closed unit disk capture a global minimizer?

5. Why is it useful to find one point inside the disk where $f < 0$?

Continuity and coercive estimate.

Show f is continuous as a sum/composition of continuous functions:

Use $\sin(x - y) \geq -1$ to prove $f(x, y) \geq$

Conclude that if $x^2 + y^2 \geq 1$, then $f(x, y) \geq$

Find a compact region where the minimum occurs.

Choose $p = (-1/2, 1/2)$. Compute $f(p) =$

Explain why $f(p) < 0$:

Let $K = \{(x, y) : x^2 + y^2 \leq 1\}$. Explain why K is compact:

Apply the extreme value theorem to $f|_K$:

Compare points inside K and outside K to prove the minimizer on K is global:

Use this page for the setup and compact restriction.

Use this page to prove the minimum is global.

Student-Derived Solution (No official solution available yet)

First, f is continuous on $\mathbb{R}^2$. The map $(x, y) \mapsto x^2 + y^2$ is continuous, the map $(x, y) \mapsto x - y$ is continuous, and $\sin$ is continuous by the allowed calculus facts. Hence $(x, y) \mapsto \sin(x - y)$ is continuous, and so f is continuous. Since $\sin(x - y) \geq -1$, we have

$$f(x, y) = x^2 + y^2 + \sin(x - y) \geq x^2 + y^2 - 1.$$

In particular, if $x^2 + y^2 \geq 1$, then $f(x, y) \geq 0$.

Now consider

$$p = \left(-\frac{1}{2}, \frac{1}{2}\right).$$

Then

$$f(p) = \frac{1}{4} + \frac{1}{4} + \sin(-1) = \frac{1}{2} - \sin(1) < 0.$$

Also p lies in the closed unit disk

$$K = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}.$$

The set K is closed and bounded in $\mathbb{R}^2$, hence compact. By the extreme value theorem, since f is continuous, there exists $q \in K$ such that

$$f(q) \leq f(z) \quad \text{for all } z \in K.$$

We claim q is a global minimizer on $\mathbb{R}^2$. If $z \in K$, this is true by the choice of q . If $z \notin K$, then $z = (x, y)$ satisfies $x^2 + y^2 > 1$, so $f(z) \geq 0$. But

$$f(q) \leq f(p) < 0 \leq f(z).$$

Thus $f(q) \leq f(z)$ for every $z \in \mathbb{R}^2$. Therefore f has a global minimum.

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 6

Problem Statement

Homework 12, Problem 6

Let X and Y be metric spaces and $f : X \rightarrow Y$ a function. Prove that if X is compact and f is continuous and bijective, then the inverse function $f^{-1} : Y \rightarrow X$ is also continuous. (Hint: It suffices to show that the preimage of any closed subset of X under f^{-1} is closed in Y .)

Topic: _____

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

1. Which closed-set criterion for continuity is suggested by the hint?

2. If $C \subseteq X$ is closed, why is C compact?

3. Why is $f(C)$ compact in Y ?

4. Why is $f(C)$ closed in Y ?

5. Why is $(f^{-1})^{-1}(C) = f(C)$ when f is bijective?

Use the closed-set criterion for f^{-1} .

Let $C \subseteq X$ be closed. Since X is compact, conclude C is:

Since f is continuous, conclude $f(C)$ is:

Since Y is a metric space, compact subsets of Y are:

Use bijectivity to compute $(f^{-1})^{-1}(C) =$

Conclude the preimage under f^{-1} of every closed set $C \subseteq X$ is closed in Y :

Apply the closed-set criterion to finish:

Use this page for the complete proof.

Student-Derived Solution (No official solution available yet)

We prove that $f^{-1} : Y \rightarrow X$ is continuous using the closed-set criterion.

Let $C \subseteq X$ be closed. Since X is compact and C is closed in X , C is compact. Since $f : X \rightarrow Y$ is continuous, the image of a compact set under f is compact. Therefore $f(C)$ is compact in Y .

Because Y is a metric space, compact subsets of Y are closed. Hence $f(C)$ is closed in Y .

Since f is bijective,

$$(f^{-1})^{-1}(C) = f(C).$$

Thus the preimage under f^{-1} of every closed subset $C \subseteq X$ is closed in Y . By the closed-set criterion for continuity, f^{-1} is continuous.

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:
